



JEE MAIN 2026

Percentile Analyzer Data Feed & Product Spec

A practical reference for building a StudyFAM marks -> estimated percentile -> estimated AIR experience, using the latest completed JEE Main Paper 1 cycle as the baseline.

CORE PRINCIPLE

Do not show an exact percentile or exact AIR from marks alone. Use a range, because NTA computes percentile shift-wise and normalises across shifts.

Reference date: 03 October 2026 | Latest official cycle: JEE Main 2026

1. Current JEE Main 2026 snapshot

Official NTA facts that should anchor the website model

15,38,468 Unique Paper 1 candidates appeared in 2026	16,04,854 Unique candidates registered across both sessions	75 / 300 Questions / maximum marks in Paper 1	2 Sessions; best NTA score used for ranking
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What the official 2026 structure means for your analyzer

JEE Main 2026 Paper 1 (B.E./B.Tech.) had Mathematics, Physics and Chemistry with 20 MCQs + 5 numerical-value questions per subject: 75 questions and 300 marks in total. Both MCQs and numerical-value questions carried +4 for a correct answer, -1 for an incorrect answer, and 0 for an unanswered response.

NTA conducted JEE Main 2026 in two sessions. For candidates who appeared in both sessions, the better NTA score was used for the merit list/ranking. The final 2026 result reports 15,38,468 unique Paper 1 candidates appeared across January and April.

FACT	2026 reference	Why it matters for StudyFAM
Paper 1	B.E./B.Tech.; PCM	Use only Paper 1 data for the B.Tech analyzer.
Maximum score	300	Marks input should be constrained to 0-300.
Questions	75	75-question mock/score assumptions align with 2026 structure.
Scoring	+4 / -1 / 0	Raw marks are not the same thing as percentile.
Sessions	2	Allow Session 1 / Session 2 / Both in advanced mode.
Final candidate pool	15,38,468 appeared	Useful baseline for converting percentile into an approximate AIR range.

Product rule: the website should display the 2026 values as a historical baseline and explicitly state that a future 2027 result can differ. The 2027 model should be refreshed when NTA releases actual candidate counts and score distributions.

2. How NTA percentile actually works

Use this logic in the product explanation and help tooltip

NTA defines the percentile score from the candidate's shift-wise relative performance. The percentile represents the percentage of candidates in that shift who scored equal to or below the candidate. NTA then merges the shift-wise percentile scores across the multi-shift examination for preparation of the result.

$$\text{Percentile} = 100 \times (\text{number of candidates in the shift with raw score} \leq \text{candidate score}) / (\text{total candidates who appeared in that shift})$$

Why marks alone are not enough

- Different shifts can have different difficulty levels. NTA explicitly states that this is possible and uses percentile-based normalisation to reduce advantage or disadvantage caused by shift difficulty.
- Two candidates with the same raw marks can therefore have different estimated percentiles depending on their shifts.
- For this reason, the quickest StudyFAM calculator should return a percentile range; the advanced calculator can ask for session/date/shift and use a narrower historical curve.

Recommended calculation pipeline

STEP	ACTION	OUTPUT	IMPLEMENTATION NOTE
1	Input raw marks	0-300	Validate numeric range
2	Select model	Quick / Advanced	Quick = pooled 2026 baseline; Advanced = shift-aware
3	Estimate percentile	Range first	Use historical curve / interpolation
4	Estimate AIR	Range	Map percentile to 2026 candidate pool
5	Show confidence	Low / Moderate / High	Depends on shift information + model coverage

Never label the estimate as NTA score. Use wording such as "Estimated percentile" or "Likely percentile range". The NTA percentile is the official normalised score; StudyFAM is estimating it from external trend data.

3. StudyFAM pooled 2026 marks -> percentile -> AIR baseline

Use as a product baseline, not as an official NTA conversion table

The following ranges are intentionally broad. They are designed for a quick marks-only calculator and reflect the spread seen in 2026 public shift analyses. At high percentiles, small changes in percentile can correspond to large changes in rank, so the analyzer should prefer ranges to a single exact rank.

Marks / 300	Estimated percentile	Approx. AIR (15,38,468 pool)
271-300	99.99-100.00	1-154
250-270	99.95-99.99	154-769
230-249	99.87-99.95	769-2,000
210-229	99.69-99.87	2,000-4,769
190-209	99.41-99.69	4,769-9,077
170-189	98.88-99.41	9,077-17,231
160-169	98.53-98.88	17,231-22,615
150-159	98.09-98.53	22,615-29,385
140-149	97.54-98.09	29,385-37,846
130-139	96.88-97.54	37,846-48,000
120-129	96.07-96.88	48,000-60,462
110-119	95.06-96.07	60,462-76,000
100-109	93.80-95.06	76,000-95,385
90-99	92.22-93.80	95,385-1,19,693
80-89	90.28-92.22	1,19,693-1,49,539
70-79	87.52-90.28	1,49,539-1,92,001
60-69	83.89-87.52	1,92,001-2,47,847
40-59	69.58-83.89	2,47,847-4,68,002
20-39	36.58-69.58	4,68,002-9,75,696
0-19	5.71-36.58	9,75,696-14,50,621

AIR calculation baseline: for a simple pooled model, use $\text{AIR approx} = (100 - \text{percentile}) \times 15,38,468 / 100$, then clamp to a sensible range and round. The real merit list is determined from NTA scores and tie rules, so this should remain an estimate.

4. 2026 shift effect: why the advanced analyzer matters

Examples from public 2026 shift-wise marks vs percentile analyses

Public 2026 shift analyses show meaningful variation in the raw marks associated with the same percentile. Careers360's 2026 shift table, for example, reports the following 99th-percentile mark ranges in April and point estimates in January.

April shift	Marks at about 99 percentile	January shift	Marks at about 99 percentile
2 Apr S1	176-180	21 Jan S1	167+
2 Apr S2	181-185	21 Jan S2	171+
4 Apr S1	186-190	22 Jan S1	158+
4 Apr S2	191-195	22 Jan S2	155+
5 Apr S1	171-175	23 Jan S1	158+
5 Apr S2	166-170	23 Jan S2	163+
6 Apr S1	151-155	24 Jan S1	162+
6 Apr S2	161-165	24 Jan S2	160+
8 Apr S2	156-160	28 Jan S1	161+
		28 Jan S2	162+

Product implication

The same 180 raw marks could land in materially different percentile zones depending on shift difficulty. Therefore, a user-facing result should show a band such as "Estimated percentile: 98.9-99.4" rather than a false precision such as "99.17 percentile" when no shift is supplied.

QUICK MODE	ADVANCED MODE
Input: 180 marks	Input: 180 marks + session/date/shift
Output: wider percentile and AIR range	Output: narrower percentile and AIR range using shift-specific data
Confidence: Moderate or Low	Confidence: Moderate or High when coverage is strong

5. Recommended StudyFAM implementation

Suggested data model and UI output for the website

Recommended user inputs

FIELD	VALUE / RULE
Mode	Quick or Advanced
Raw marks	0-300
Session	Optional in Quick; required in Advanced
Exam date / shift	Optional in Quick; used in Advanced when known

Example JSON feed structure

```
{
  "year": 2026,
  "exam": "JEE Main Paper 1",
  "max_marks": 300,
  "candidate_pool": 1538468,
  "mode": "pooled_baseline",
  "bands": [
    { "min_marks": 170, "max_marks": 189,
      "percentile_min": 98.88, "percentile_max": 99.41 }
  ]
}
```

Recommended result card

YOUR JEE MAIN ANALYSIS
180 / 300 marks
Estimated percentile: 98.9-99.4
Estimated AIR: 9,000-17,500
Confidence: Moderate
Based on 2026 public shift/trend data. Actual percentile and AIR depend on the candidate's shift, NTA normalisation and final candidate distribution.

6. Source notes & launch checklist

Keep these references attached to the analyzer logic for transparency

Primary sources

SOURCE	USE
NTA JEE Main 2026 Information Bulletin https://jeemain.nta.nic.in/document/ib-2026/	Official exam scheme, marking, two-session structure, and percentile normalisation.
NTA JEE Main 2026 final result notice https://www.nta.ac.in/Download/Notice/Notice_20260420194021.pdf	Official candidate counts and final 2026 Paper 1 NTA Score information.
JEE Main 2026 official documents page https://jeemain.nta.nic.in/documents/	Official index of 2026 result, answer-key and session documents.
Careers360 - JEE Main 2026 marks vs percentile https://engineering.careers360.com/articles/jee-main-marks-vs-percentile	Public shift-wise 2026 marks vs percentile trend tables used to illustrate shift variation.

Launch checklist

- ☐ Constrain marks input to 0-300 for Paper 1.
- ☐ Show "Estimated" everywhere; never imply that the calculator is an NTA official tool.
- ☐ Return a percentile range and AIR range in Quick mode.
- ☐ Add session/date/shift as optional advanced inputs for a narrower estimate.
- ☐ Keep 2026 as the historical baseline and version the data feed so it can be replaced with 2027 evidence later.
- ☐ Store the source date and confidence level alongside every estimate.

Important: NTA does not publish an official marks-to-percentile conversion table. The marks-to-percentile ranges in this document are therefore a StudyFAM estimation baseline assembled from publicly available 2026 trend analyses. The official NTA percentile remains the authoritative result.

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